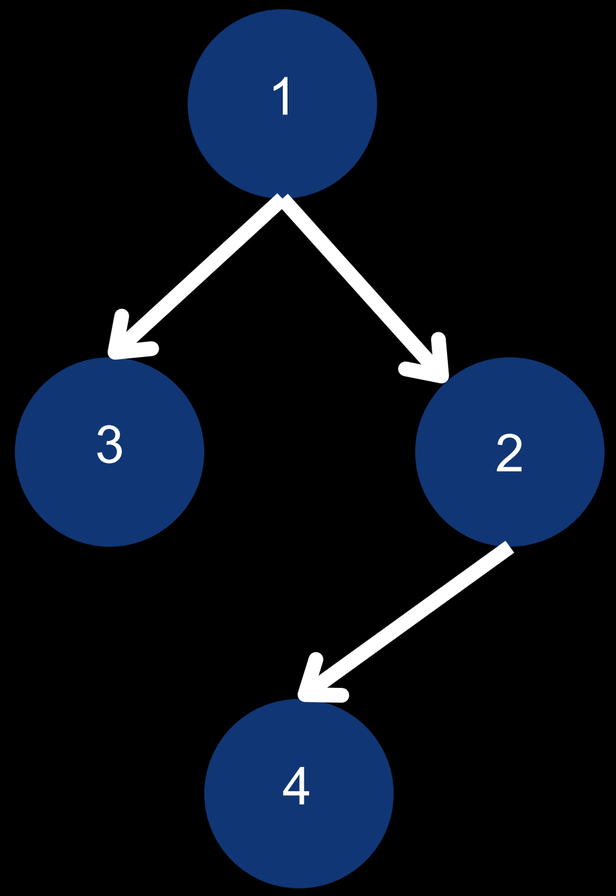
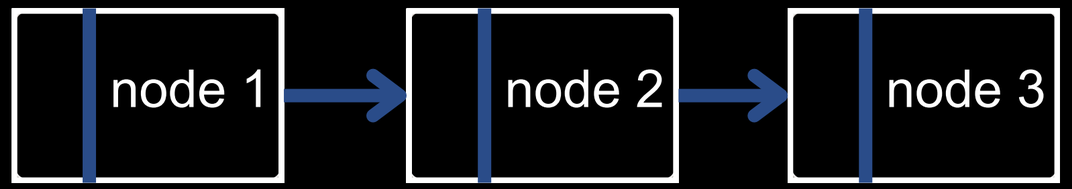
DATA STRUCTURE





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Searching

Linear Search

Linear Search, also known as Sequential Search, is the simplest searching algorithm. It traverses a list or array from the beginning to the end, comparing each element with the target value until a match is found or the list ends.

How Does Linear Search Work?

Start from the first element of the array.

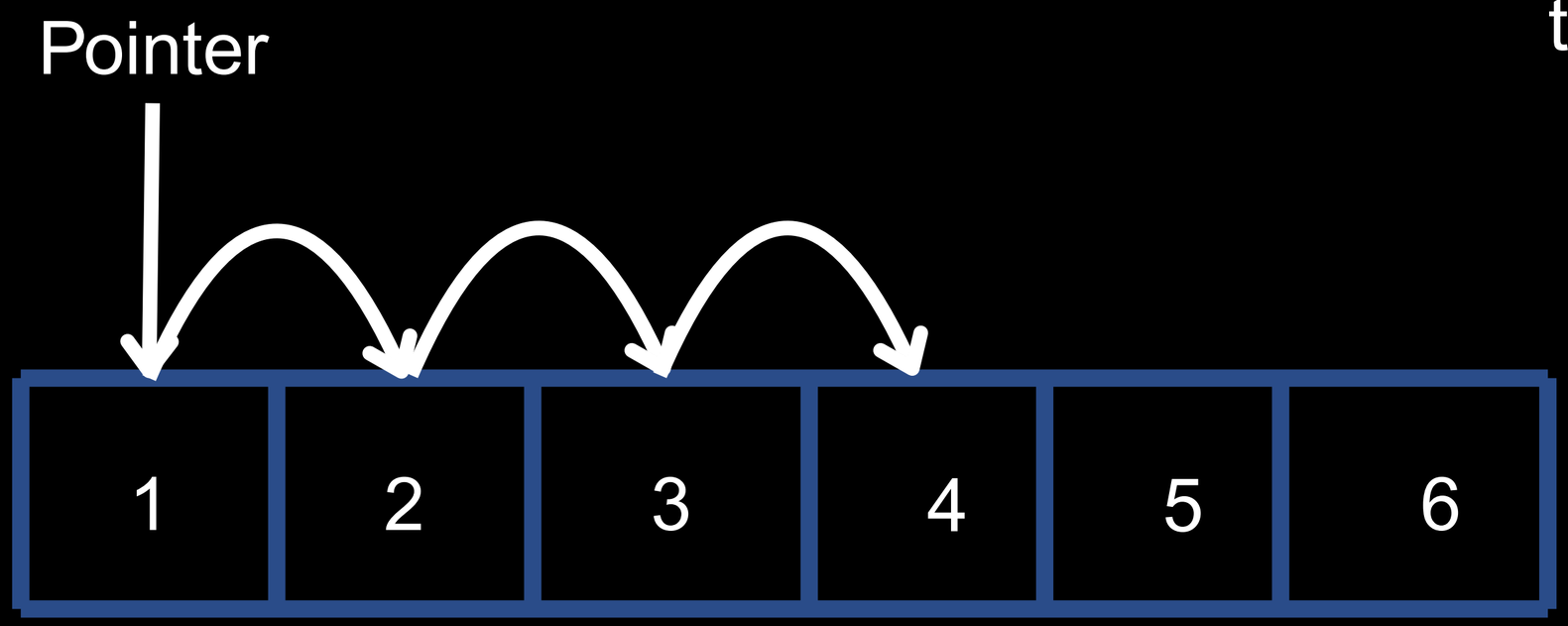
Compare the current element with the target value.

If it matches, return the index of the element.

If it doesn’t match, move to the next element.

Repeat steps 2–4 until the target is found or the end of the array is reached.

traget = 4



When to Use Linear Search

Unsorted Data: Linear Search doesn’t require the data to be sorted.

Small Data Sets: Efficient for small arrays where the overhead of more complex algorithms isn’t justified.

Simplicity: Easy to implement and understand, making it suitable for beginners.

Binary Search

Binary Search is a highly efficient searching algorithm that works on sorted arrays. It repeatedly divides the search interval in half, eliminating half of the remaining elements each time, until the target value is found or the interval is empty.

How Does Binary Search Work?

Prerequisite: The array must be sorted in ascending or descending order.

Determine the middle element of the array.

Compare the middle element with the target value.

If it matches, return the index.

If the target is smaller, repeat the search on the left half.

If the target is larger, repeat the search on the right half.

Continue the process until the target is found or the search interval is empty.

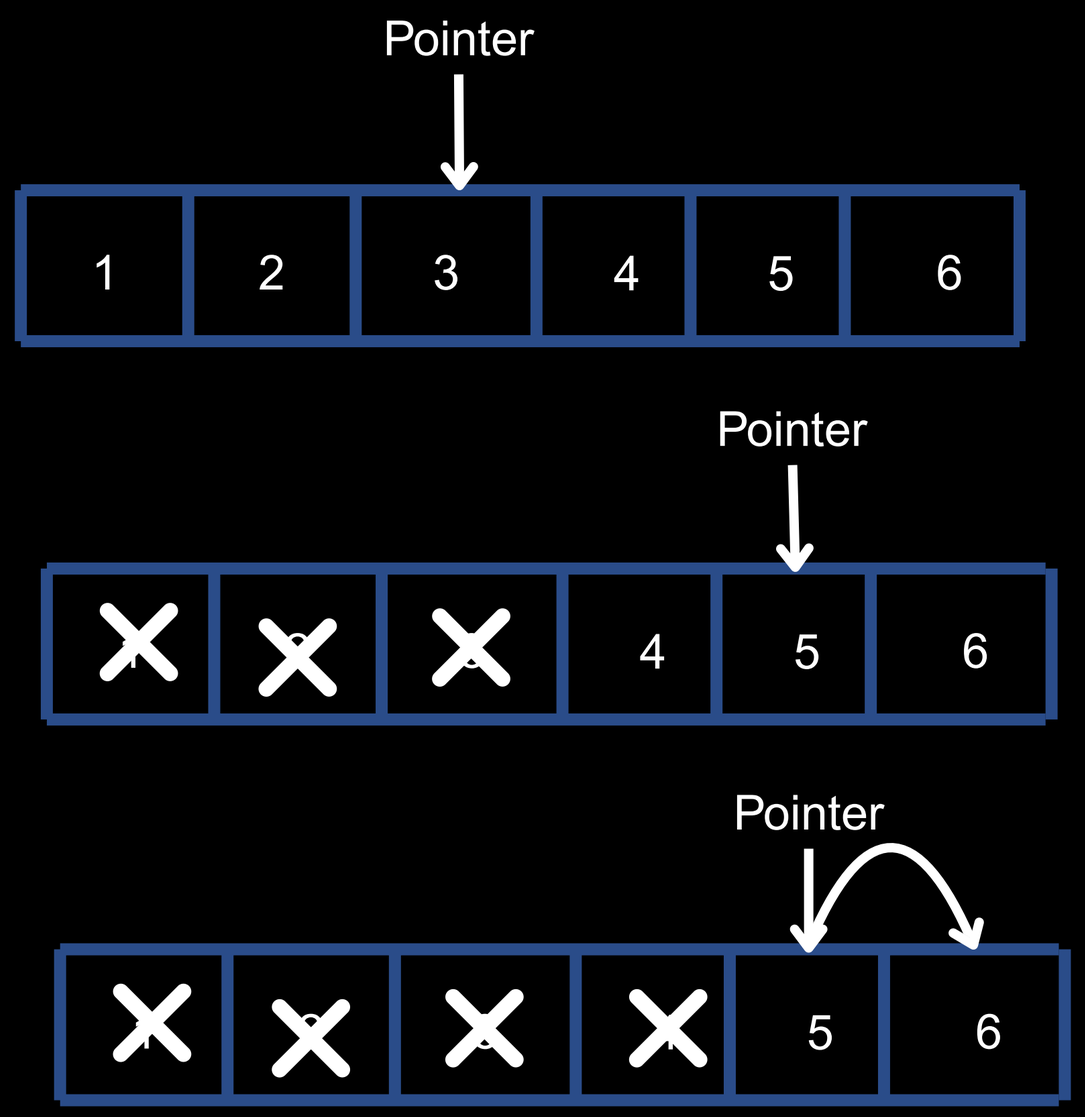
When to Use Binary Search

Sorted Data: Binary Search requires the array to be sorted.

Large Data Sets: Highly efficient for large arrays, reducing the number of comparisons significantly.

Performance-Critical Applications: Ideal when performance is a priority, and the overhead of sorting (if needed) is manageable.

traget = 6



Thanks

All code and examples can be found on GitHub https://github.com/ESSAMMOHAMED1